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Note from author

These notes are mostly based on the syllabus and past papers

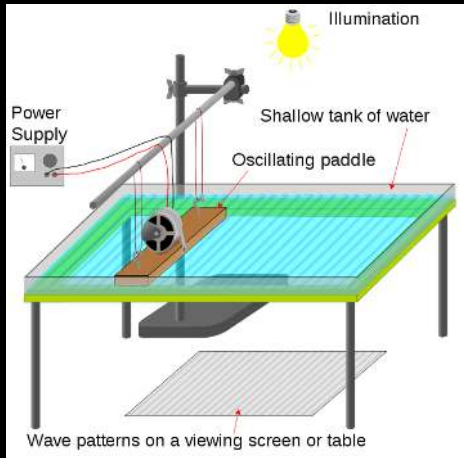
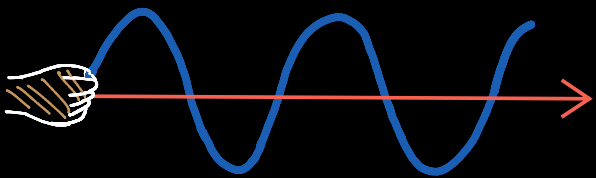
Fair warning: they might look short (2–3 pages), but I've condensed everything down to keep the important stuff while removing the extra yap. That said, I wouldn't recommend using these as quick revision notes, but think of them as detailed notes covering almost everything

Spotted a mistake? Suggestions?

I'd love to hear your feedback. Contact us at: learnmates.share@gmail.com

Progressive wave

A progressive wave (also called a travelling wave) is a disturbance that transfers energy from one point to another without transferring the material medium itself.



Using a ripple tank to observe a wave:

- Light source placed above the ripple tank
- White screen placed below the tank
- Light shines through water onto screen to create wave shadow
- Bright areas on screen = wave crests and Dark areas on screen = wave troughs
- Use a stroboscope to "freeze" the waves for measurement and Adjust strobe frequency until waves appear stationary
- Use a camera to photograph the pattern for analysis
- Can measure wavelength directly from the frozen image

Describing waves

• Wavelength (λ): minimum distance between two points moving in phase (e.g. from one crest to the next, or one trough to the next). **Measured in metres (m).**

• Displacement (x): How far a point on the wave (or a particle in the medium) is from its resting (equilibrium) position at any given time. It can be positive or negative.

• Amplitude (A): The maximum **magnitude** of the displacement from the equilibrium position. A larger amplitude means the wave carries more energy. **Measured in metres (m).**

• Period (T): The time taken for one complete wave (oscillation) to pass a fixed point. **Measured in seconds (s).**

• Frequency (f): The number of complete waves (oscillations) passing a **fixed point** on the wave per unit time. **Measured in Hertz (Hz),** where 1 Hz = 1 wave per second.

The frequency and period are reciprocals: $f = \frac{1}{T}$

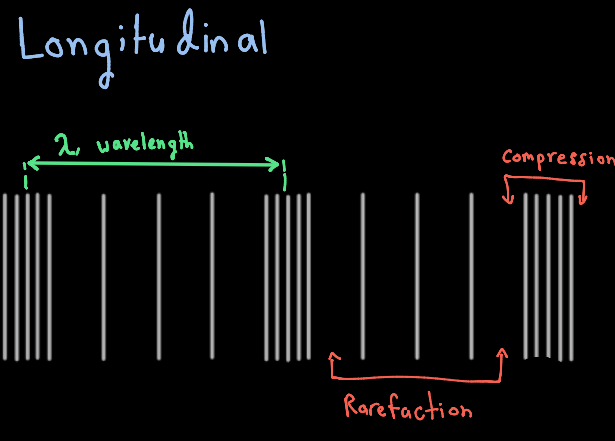
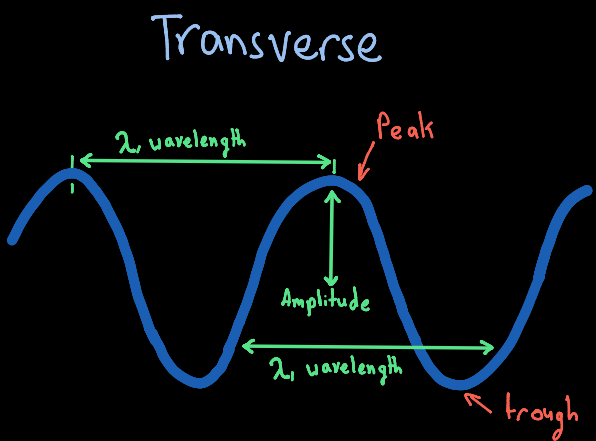
• Speed: The speed at which the wave's energy propagates (travels) through the medium. **Measured in metres per second (m/s).**

$$\text{Speed} : \frac{d}{t} = \frac{n \times \lambda}{t} = \frac{\lambda}{\frac{t}{n}} = f \lambda$$

$\frac{1}{T} = f$

Transverse Vs. Longitudinal Waves

Feature	Transverse Wave	Longitudinal Wave
Direction of Vibration	Perpendicular (at right angles) to the direction of energy transfer.	Parallel to the direction of energy transfer.
Appearance	Has peaks (crests) and lowest points (troughs).	Consists of compressions (where particles are closest) and rarefactions (where particles are most spread out).
Can it travel in a vacuum?	Yes (e.g., light, all EM waves).	No (requires a medium, e.g., sound needs air/water/solids).
Examples	Waves on a string, water waves (surface), electromagnetic waves.	Sound waves, seismic P-waves, waves in a slinky when pushed and pulled.



Intensity of Light

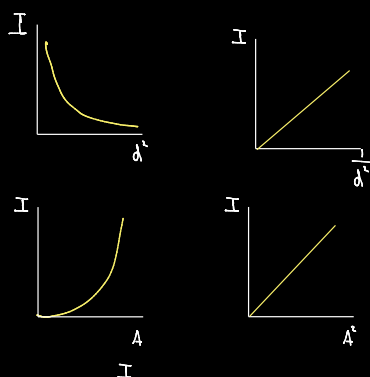
Intensity is the power transferred per unit area: $I = \frac{P}{A}$

Key Proportionality Relationships:

Intensity and distance: $I \propto \frac{1}{d^2} \Rightarrow \frac{I_1}{I_2} = \left(\frac{d_2}{d_1}\right)^2$

Intensity and Amplitude: $I \propto A^2 \Rightarrow \frac{I_1}{I_2} = \left(\frac{A_1}{A_2}\right)^2$

Intensity and frequency: $I \propto f^2 \Rightarrow \frac{I_1}{I_2} = \left(\frac{f_1}{f_2}\right)^2$



Cathode-Ray Oscilloscope (CRO)

A Cathode-Ray Oscilloscope (CRO) is an electronic instrument that displays voltage signals as a waveform on a screen. For wave experiments, we connect a signal source (like a microphone receiving sound waves, or a signal generator producing waves) to the input.

The CRO allows us to visualise an electrical signal that varies with time. By analysing the shape on the screen, we can determine two crucial properties of a wave: its **amplitude** and its **frequency** (or period).

The screen has a grid of squares called divisions (often abbreviated as div). These divisions are used for measurement.

Voltage-Gain (Y-Gain) and Time-Base

Y-Gain: Controls the vertical scale. It determines how many volts are represented by each vertical division on the grid.

Affects: Amplitude (voltage)

Unit: Volts per Division (V/div)

Determining Amplitude: Vertical peak to trough distance / 2 $\times$ Voltage gain **OR**

Vertical center to trough distance (crest to trough) $\times$ Voltage gain

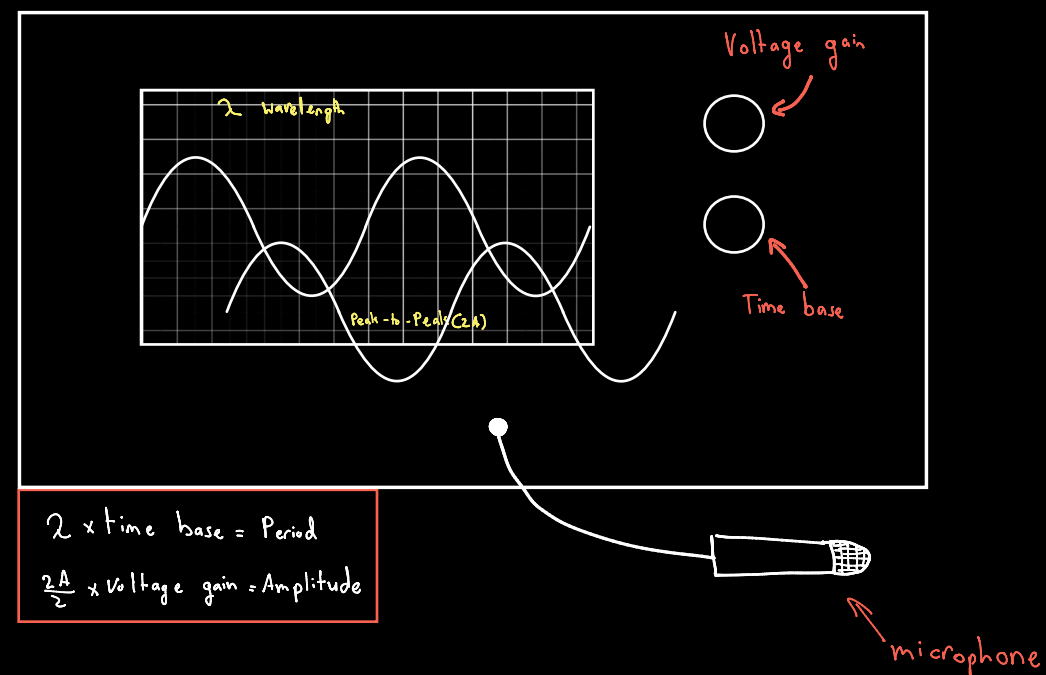
Time-Base: Controls the horizontal scale. It determines how much time is represented by each horizontal division on the grid.

Affects: Period / Frequency (Time)

Unit: Time per Division (s/div)

Determining period: Horizontal peak to peak distance (wavelength) $\times$ time base

Frequency = 1 / period =



$$\lambda \times \text{time base} = \text{Period}$$

$$\frac{\lambda A}{2} \times \text{Voltage gain} = \text{Amplitude}$$

Phase

Phase of a particles: a way of describing the current position of a point on a wave within its cycle. It tells us how far through the oscillation a particular particle (or point on the wave) has progressed.

Phase is measured in:

Degrees ($^\circ$) – where one complete cycle = 360°

Radians (rad) – where one complete cycle = 2π rad

Phase difference

Phase difference: the amount by which one wave (or point on a wave) leads or lags behind another wave (or point). It tells us how "out of step" two waves are.

it is difference between:

Two points on the same wave (e.g. two particles in a medium)

Two different waves of the same frequency: choose equivalent points on the two waves to compare (crest-crest)

starting point	0° (or 360°)
Quarter of the way through ($\frac{1}{4}$)	90° ($\pi/2$ rad)
Halfway ($\frac{1}{2}$)	180° (π rad)
Three-quarters of the way ($\frac{3}{4}$)	270° ($3\pi/2$ rad)
Back to starting position (λ)	360° (2π rad)

In Phase (Phase Difference = 0° or 360° , or 2π rad):

They have the **same displacement and are moving in the same direction.**

Path difference = $n\lambda$ where $n=0,1,2,3...$ (they are separated by a whole number of wavelengths/freq)

Phase difference = 0 (or $360^\circ \times n$)

Out of phase:

They are not always moving in the same direction (not in phase)

Anti-Phase (Phase Difference = 180° or π rad):

Special type of "out of phase" where they are moving in opposite directions.

Path difference = $(n+1/2)\lambda$ where $n=0,1,2,3...$

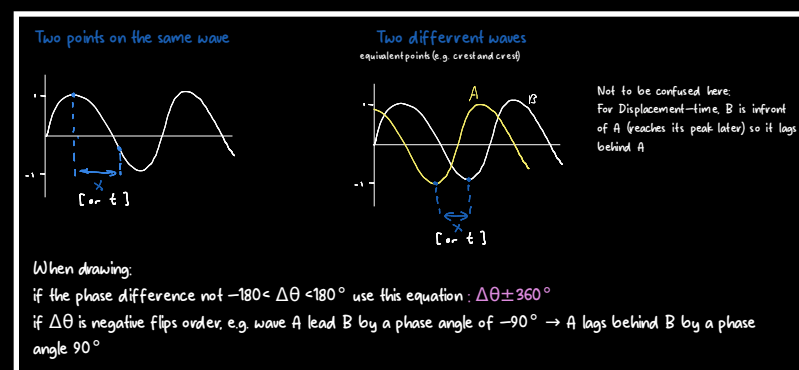
Phase difference = 180° (or π rad)

$$\frac{\Delta \theta}{360^\circ} = \frac{\Delta t}{T} = \frac{\Delta x}{\lambda}$$

$\Delta \theta$ = phase difference

Δx = distance difference (for displacement-distance)

Δt = period/time difference (for displacement-time)



Coherence

Two waves are considered coherent if:

- They have a **constant phase difference**
- They have the same frequency

Ordinary light sources (like filament lamps, fluorescent tubes, or the sun): do not produce coherent waves. This is due to two main reasons:

Multiple Frequencies: They emit light at **many different frequencies** (wavelengths/colours).

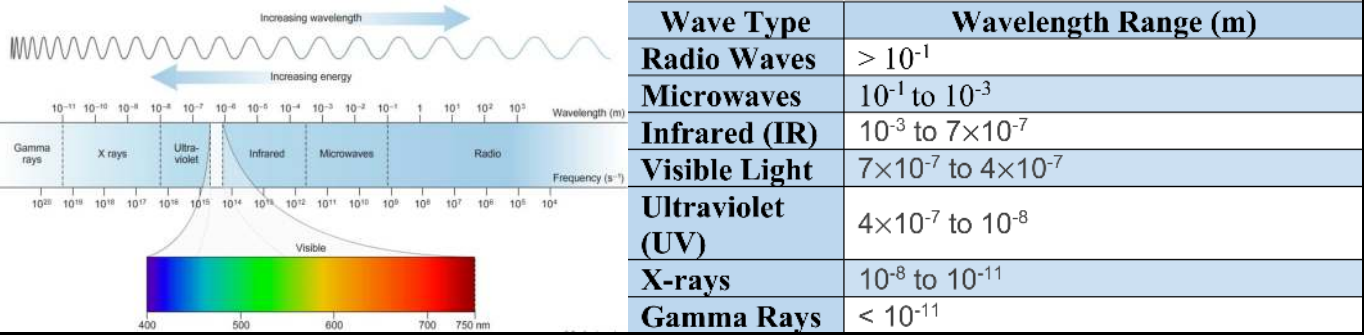
Random Phase: The light is emitted by billions of independent atoms. Each atom emits a short wave of light before stopping and **starting again at a random phase**.

Lasers: unlike ordinary sources, lasers produce coherent waves as they emit single frequency waves and have no random phase changes.

Electromagnetic waves

Properties:

- They are transverse waves meaning their oscillation is perpendicular to energy transfer
- They can travel through a vacuum (do not need a medium) unlike mechanical waves
- In a vacuum they have a constant speed c (3×10^8)
- They can be reflected, refracted, diffracted and polarized



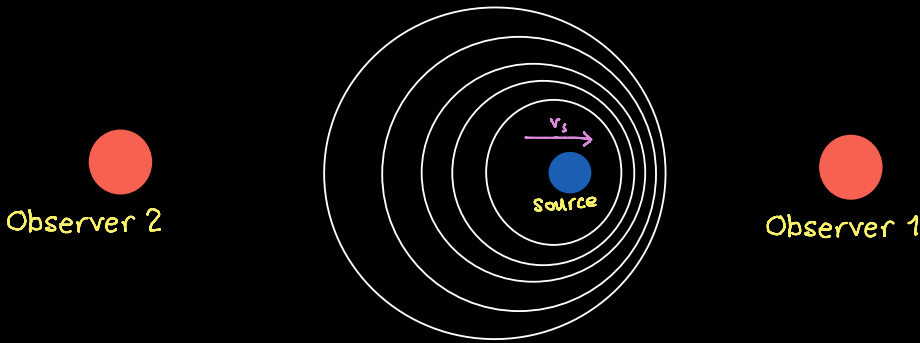
Doppler effect

The Doppler Effect is when the observed frequency is different to source frequency when source moves relative to observer.

$$f_o = \frac{f_s v}{(v \pm v_s)}$$

f_s = Source frequency
 f_o = Observed frequency
 v = Velocity of waves
 v_s = Velocity of source

Example

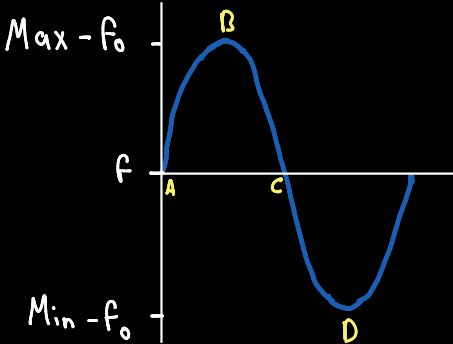
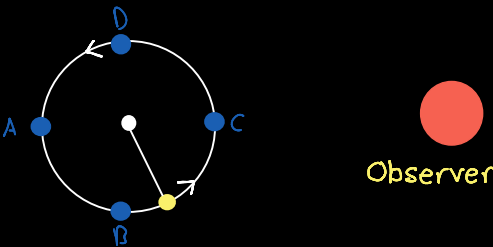


	Wavefront	Wavelength	Frequency	Formula
Observer 1 (Source moving towards it)	Compressed	Decreases	Increases	$f_o = \frac{f_s \times v}{v - v_s}$
Observer 2 (Source moving away from it)	Stretched out	Increases	Decreases	$f_o = \frac{f_s \times v}{v + v_s}$

Doppler effect in circular motion

When an object moves in a circle, its speed relative to an observer constantly changes. This causes the observed frequency to increase and decrease in a repeating pattern.

- Moving towards observer $\rightarrow$ frequency increases.
- Moving away from observer $\rightarrow$ frequency decreases.
- Moving sideways $\rightarrow$ frequency unchanged.



Polarization

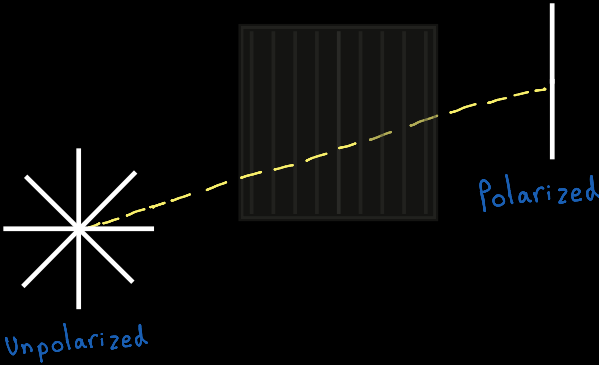
Polarization is the process that results in oscillations in a single direction, perpendicular to the direction of wave propagation.

Transverse waves are waves that vibrate perpendicular to the direction of energy transfer.
Can be polarized.

Longitudinal waves are waves that vibrate parallel to the direction of energy transfer.
Cannot be polarized

In **unpolarized** waves vibrations occur in **many different planes**, all perpendicular to the direction of travel. Sunlight and light from a bulb are examples of unpolarized light. When **polarized** vibrations are **confined to a single plane** only, and the plane at which the polarized wave vibrates is called the **plane of polarization**.

Transverse waves can be polarized using a **polarizing filter**, which acts like a series of parallel slots. It allows only vibrations aligned with its axis to pass through while absorbing the rest. and the axis along which the waves pass through is called the **transmission axis**



Intensity and Polarization

1. Unpolarized Light $\rightarrow$ First Polarizer

When unpolarized light hits a polarizing filter:

- The filter only allows vibrations aligned with its transmission axis to pass.
- Unpolarized light vibrates equally in all directions.
- Only one direction (the component parallel to the axis) gets through.

Result: Intensity is halved
 $I = \frac{1}{2} I_o$

I_o = initial intensity of unpolarized light
 I = intensity after first polarizer

Polarized Light $\rightarrow$ Second Polarizer

When already polarized light hits a second polarizer the transmitted intensity depends on the angle θ between the transmission axes of polarizer and initial plane of polarization of wave.

This is given by Malus's Law:

$$I = I_o \times \cos^2 \theta$$

Where:

I_o = initial intensity of polarized light
 I = intensity after second polarizer
 θ = angle between the transmission axes of polarizer and initial plane of polarization of wave.

